

Igor Simões

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ABOUT

Software engineer with deep expertise in backend architecture, agentic automation, and large scale data pipelines.
Proven record of improving system resilience, optimizing cloud costs, and scaling distributed systems to handle millions of real time events.
Product minded, leading the full cycle of AI platforms and cloud native systems under startup pressure.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, C/C++
Frameworks & Libraries: FastAPI, Django REST, Flask, Node.js, Next.js, React, Flutter, Celery, GraphQL
Databases & Search: PostgreSQL, Redis, Neo4j, Qdrant, Weaviate, MongoDB, DynamoDB, Mem0
AI & Machine Learning: PyTorch, Hugging Face, LangChain, NumPy, Pandas
Cloud & DevOps: AWS (EC2, ECS, S3, RDS, CloudWatch, Athena), GCP, DigitalOcean, Terraform, Docker, Kubernetes, Kafka, Nginx, Traefik, Linux, Git
Developer Tools: Cursor, Claude, Antigravity, Context Engineering, Harness Engineering

EXPERIENCE

Peacore

Cofounder & Director of Technology

Software House / Consulting (Apr/2025 - Present)

- Restructured internal processes for nonprofit organizations, including IEPÉ, by delivering custom systems, relational data modeling, and integrations on **AWS** and **Supabase** with a streamlined automated deployment cycle.
- Grew the firm by defining product vision and company positioning, structuring technical and commercial proposals, prospecting clients, and acting as a strategic technology partner.
- Built the delivery function by leading interns, owning architecture decisions, and enforcing clean architecture and software engineering best practices across projects.

Blumi

- Revamped an AI built platform into a scalable production system by designing a new architecture and enforcing engineering processes such as Gitflow and Hexagonal Architecture using **Next.js** and **Nest.js**.
- Built and scaled a fault tolerant messaging platform for transactional communications, implementing strategic error catching, number rotation, and retry and backoff strategies using **Python**.
- Architected an event driven pipeline for logs, metrics, and customer insights using **Golang** and **Node.js** event streams, and delivered analytics dashboards on **AWS Athena**.
- Designed the unified SSO architecture across 3 platforms, delivering the **SAML 2.0** integration for a large corporate client and a migration strategy using **Drizzle**.
- Migrated a **MongoDB** data context into **PostgreSQL** by building and managing an ETL pipeline.
- Managed the AWS architecture through **Terraform (Cognito, EC2, RDS, S3)**, provisioned separate homologation and production environments, led the engineering team, and set up product observability with **PostHog**.
- Removed the complexity of generating new customer communications from system events by architecting, provisioning, and implementing an event driven transactional email platform and its infrastructure for a startup client, using **Golang** workers, **AWS SES, SQS, CloudWatch**, and **Event Bus**, applying the **Outbox** pattern to prevent the dual write problem, and feeding the internal data platform on **Amazon QuickSight**.

Fractal Systems

Senior Software Engineer

US - Remote (May/2025 - Jun/2026)

- Multiplied concurrent task throughput up to 16x by architecting a scalable multi tenant Hexagonal Architecture for a distributed integration pipeline, using isolated and type dedicated queues with **Celery** and **Gevent**.
- Saved USD 18,000 annually and removed single vendor dependency by migrating 8 OAuth providers with canary deployments and idempotent webhooks using **Nango**.
- Prevented frontend rendering bottlenecks and secured IPC communication by optimizing the main process across a 6,000 line codebase using **Node.js**, **Electron**, and **LanceDB**.
- Guaranteed reliable offline AI execution by refactoring local LLM loading, embedding generation, and document extraction to run without external API dependencies using **Electron** and **Node.js**.
- Scaled the database horizontally by safely migrating 166GB of production data to a distributed server secured with VPC routing using **PostgreSQL** and **Traefik**.
- Scaled AI workflows under heavy load by designing agentic orchestration patterns and optimizing model pipelines through token batching and text chunking for semantic and hybrid search using **Gemma 3** and **Hugging Face**.
- Maximized hybrid search performance and reduced memory pressure by tuning database internals, CPU allocation, and segment configuration using **Qdrant** and **gRPC**.
- Prevented silent data loss across distributed workers by engineering an exponential backoff retry strategy and resolving critical OOM crashes using **PgBouncer** for connection pooling.
- Ensured SOC2 and CASA Tier 2 compliance by enforcing security policies across all endpoints through automated CI/CD security checks using **FastAPI**.
- Shipped a web app, a desktop client, and a mobile UI on a unified cross platform backend to meet tight launch deadlines using **Next.js**, **Electron**, **Flutter**, **Claude Code**, and **Cursor**, with observability via **Sentry** and **PostHog**.
- Automated a full ingestion pipeline into a **Qdrant** vector database by architecting end to end ML pipelines with **LangChain**, **Hugging Face**, and **OpenAI** for embedding search, optimizing performance under strict production memory constraints.
- Delivered features on time under tight deadlines and budgets by leading a 4 person engineering team, mentoring junior developers, and managing agile workflows in **ClickUp** with enforced **Gitflow**.
- Prevented a critical ransom DDoS attack and saved USD 5,000 in fees by implementing **Google Cloud Shield** with load balancing, rate limiting, and abuse prevention, sustaining 99.9% uptime.
- Handled business logic for millions of daily messages by engineering a backend with **FastAPI**, **Celery** (Beat/Scheduler), and **Redis** for asynchronous orchestration of large scale **SMTP** email sending, while collecting structured data for future ML models.
- Improved scalability by refactoring an AI generated legacy codebase into a clean, well structured system, optimizing schemas and integrating **Supabase** (Auth/DB) plus full cycle payments with **Stripe** and **Chargebee**.
- Automated deployment, monitoring, and performance tuning by orchestrating **GCP** infrastructure (**Cloud Run, GKE, BigQuery, Pub/Sub**) with CI/CD via **Cloud Build** and **Docker**, and added session replay and error tracking with **LogRocket** and **Sentry**.
- Reduced vendor costs and ensured technical alignment by leading technical meetings with a vendor to clarify and negotiate pricing, implementation, and capacity details.

Mission Inbox

Software Engineer, Full Stack

US Startup (Apr/2024 - May/2025)

- Prevented a projected **20% revenue loss** and retained a key enterprise client by resolving a critical API rate limiting bottleneck and tuning Linux process limits and sockets for reliability.
- Cut CI/CD deployment time by **66%** by optimizing **Kubernetes**, **Docker**, and **GitHub Actions** pipelines, while managing **DigitalOcean** infrastructure including load balancers, **NGINX** reverse proxy, SSL certificates, and database pools.
- Increased scalability by spearheading the migration of core business services from **Python** to **Golang** through modularization, reverse engineering undocumented legacy systems to ensure REST compliance and integrity.
- Boosted performance and automated critical Customer Success processes by implementing asynchronous queue processing with **Celery** and multithreading, backed by **Redis** caching.
- Restored services during downtime by maintaining UNIX email protocols (**IMAP, POP3, MIAB, Dovecot, Postfix**) through shell scripting and protocol troubleshooting.
- Accelerated client onboarding and code quality by creating API flowcharts and documentation, integrating **BetterStack** monitoring, conducting code reviews, and mentoring developers on Git best practices.
- Strengthened B2B client relationships and resolved operational conflicts by leading direct meetings to gather requirements, explain workflows, and align on solutions.

Katapult

Software Engineer, Full Stack

US FinTech (Apr/2024 - Aug/2024)

- Ensured data integrity during a critical tax data update by cross validating external API data against the internal database, using **Django ORM** and raw **PostgreSQL** queries to prevent database locks during migration.
- Enhanced financial and credit card features by modernizing a legacy object oriented codebase and mapping data workflows in close collaboration with product teams.
- Resolved underlying bugs and optimized database performance by monitoring with **Sentry** and **AWS CloudWatch** and refining complex PostgreSQL queries.
- Automated deployments across Dev, QA, and Pre prod by launching Docker containers with **Jenkins** and **AWS ECS**, managing credentials with **AWS CLI** and **AWS Vault**.
- Delivered UI solutions in frontend modules using **AngularJS**.

ARIA - AI Applications Lab, Sefaz/PB (Department of Finance)

Full Stack Developer

(2021 - 2023)

- Ensured responsiveness and scalability for the State of Paraíba by engineering a high throughput ingestion pipeline processing **1 million NFC/day** and **300k NFE/day**, handling **500 vehicle events/min** on average.
- Handled CRUD, auth, and filtering for large volumes of fiscal documents by developing APIs in **Python/Flask** for the Integrated Command and Control Center (CICC).
- Improved reliability by architecting a message broker based data validation module using **Kafka**, **GraphQL/Hasura**, **Postgres**, and **Redis** on **Docker**.
- Delivered real time fiscal dashboards using **React**, **TypeScript**, and JavaScript.

Kartado

Backend Developer

(Oct/2022 - Oct/2023)

- Optimized performance and shipped new business rules for a Highway Management SaaS by developing backend endpoints with **Python/Django REST Framework** and **Postgres**.
- Enforced reliability through unit testing (**Pytest**), linting (**Flake8, Black**), and error tracking with **Sentry**.
- Maintained platform infrastructure and assessed feature feasibility within a SCRUM team using **Jira**.

LAVID - Digital Video Applications Lab (PHI)

Frontend Developer

(Oct/2022 - Mar/2023)

- Built reusable modular UI components from Figma designs for the Paraíba State Education Enrollment System using **React**, **Redux Saga**, and **TypeScript**, ensuring best practices and component reusability.

ACADEMIC AND SOCIAL EXPERIENCE

Universidade de São Paulo (USP)

Researcher

(2026 - Present)

- Writing a research paper on reducing social inequality through optimal allocation of public financial resources, modeling the problem as Reinforcement Learning environments.

PUC-Tech (Technology League)

Cofounder / Counselor

(Apr/2025 - Apr/2026)

- Cofounded a technology league that develops interns and early career professionals through real world software projects and collaborative learning, reaching over 26 members.
- Lead initiatives focused on mentoring, project structuring, and fostering an environment where members grow both technical and professional skills.

TAIL - Technology and Artificial Intelligence League

Director

(Sep/2020 - Jul/2024)

- Delivered a Computer Vision project requested by the Fire Brigade to identify emergency elements on blueprints, owning deadlines, data labeling policies, the model deployment pipeline, and architecture using **OpenCV, YOLO, Pandas, PyTorch**, and **Pickle**.
- Made decisions on internal league matters such as project proposals and people management.

EDUCATION

[2025 – 2027] Pontifical Catholic University of São Paulo (PUC-SP), *Computer Science*
[2022 – x] UFPB, Federal University of Paraíba, *Computer Engineering*

LANGUAGES

Portuguese: Native

English: Fluent

French: Basic

Spanish: Intermediate